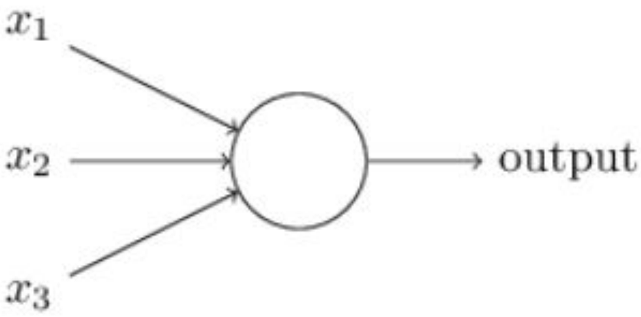
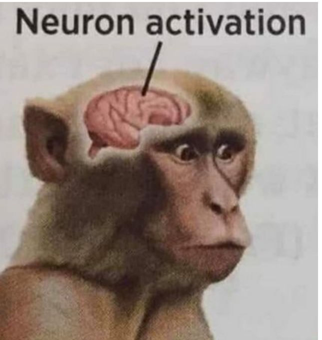
A Hitchhiker's Guide to Neural Networks

Classic Perceptron

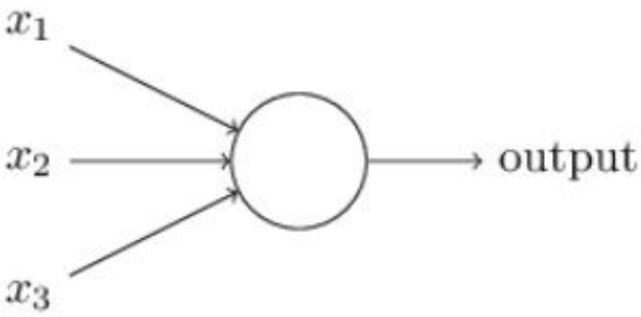


$$\text{output} = \begin{cases} 0 & \text{if } \sum_j w_j x_j \leq \text{ threshold} \\ 1 & \text{if } \sum_j w_j x_j > \text{ threshold} \end{cases}$$

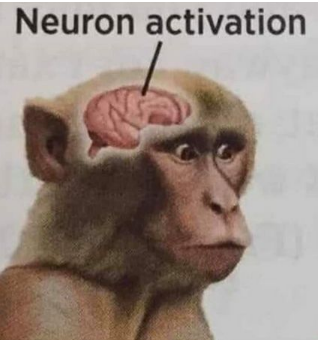


- An atomic way to *make decisions*
- Inspired by actual neurons (yes the things in your brain)

Classic Perceptron - alternate



$$\text{output} = \begin{cases} 0 & \text{if } w \cdot x + b \leq 0 \\ 1 & \text{if } w \cdot x + b > 0 \end{cases}$$



- An atomic way to *make decisions*
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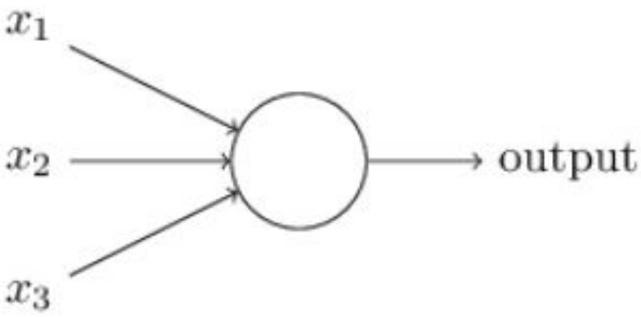
Classic Perceptron - the issue



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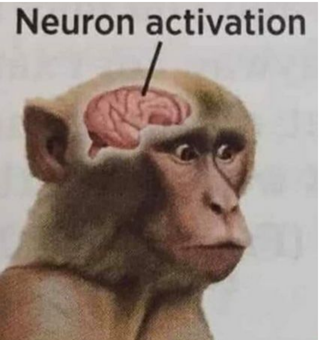
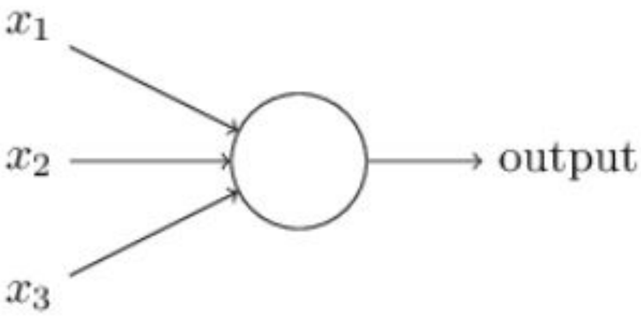


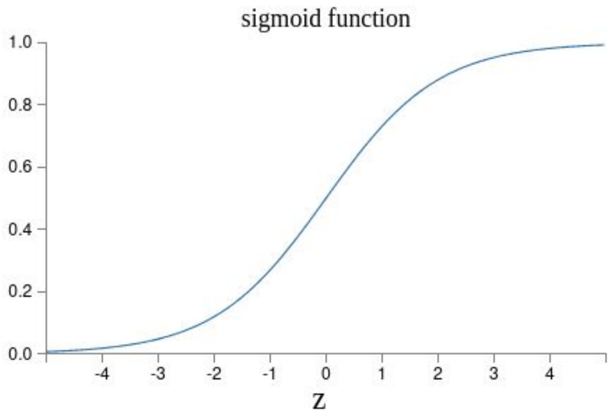
- Problem with this neuron: not differentiable
- Cannot run gradient descent

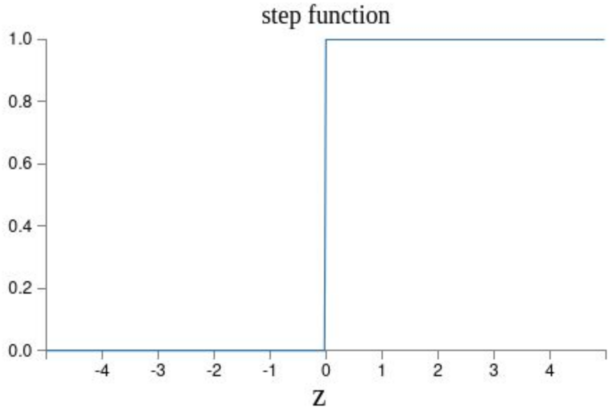
Sigmoid Perceptron



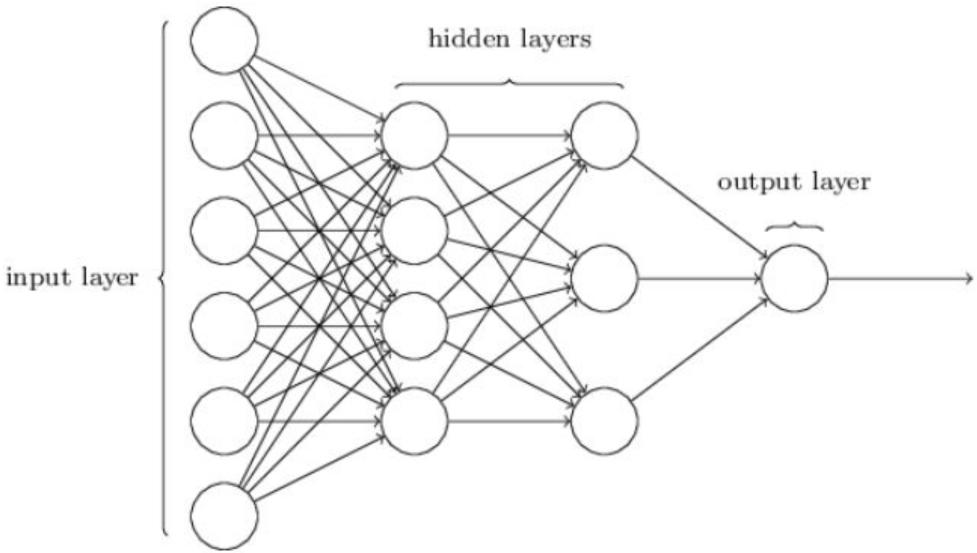
$$\sigma(z) \equiv \frac{1}{1+e^{-z}}.$$ → activation function

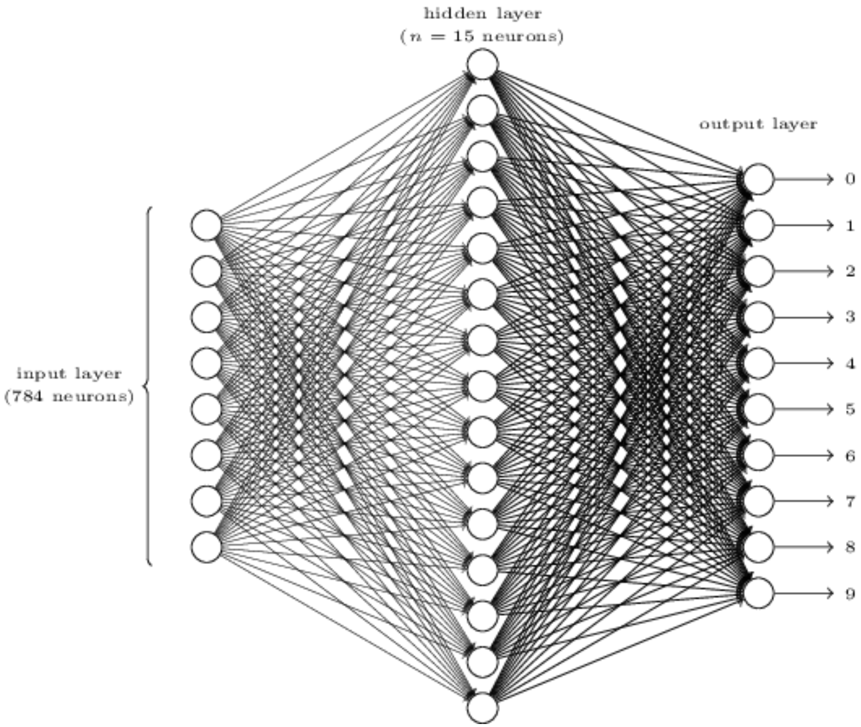
$$\text{output} = \frac{1}{1+\exp(-\sum_j w_j x_j - b)}.$$





Neural Networks (finally!)





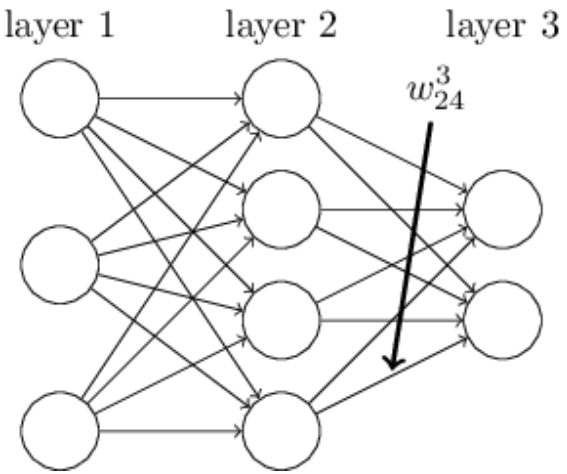
aka multilayer perceptrons (MLPs)

Universal Approximation Theorem

http://neuralnetworksanddeeplearning.com/chap4.html

Any complex function can be approximated with a single layer NN with enough neurons

Some ~~ugly~~ notation



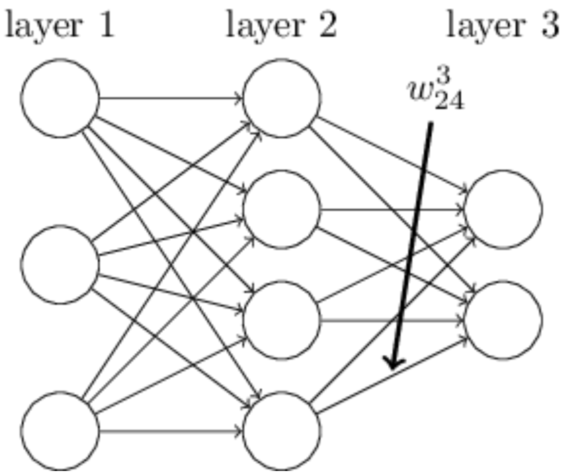
w^l_{jk} is the weight from the k^{th} neuron in the $(l-1)^{\text{th}}$ layer to the j^{th} neuron in the l^{th} layer

$$\boxed{a^l_j = \sigma\left(\sum_k w^l_{jk} a^{l-1}_k + b^l_j\right),}$$

- This will make sense very soon
- a^l_j = output of j^{th} neuron in l^{th} layer
- z^l_j = stuff inside the sigmoid

$$a^l_j = \sigma(z^l_j)$$

Some better notation



w^l_{jk} is the weight from the k^{th} neuron in the $(l-1)^{\text{th}}$ layer to the j^{th} neuron in the l^{th} layer

$$\boxed{a^l = \sigma(w^l a^{l-1} + b^l).}$$

- This notation should make sense now
- w^l = weight matrix for layer *l*
    - W_ij → weight from jth to ith neuron
- We can compute an entire batch

Cost Function

$$C = \frac{1}{2}\|y - a^L\|^2 = \frac{1}{2}\sum_j (y_j - a_j^L)^2,$$

- This is the cost for a single training example
- *L* = last layer

- Must be easily differentiable
- Cost over a batch can be written as average of individual costs (at least in this case)

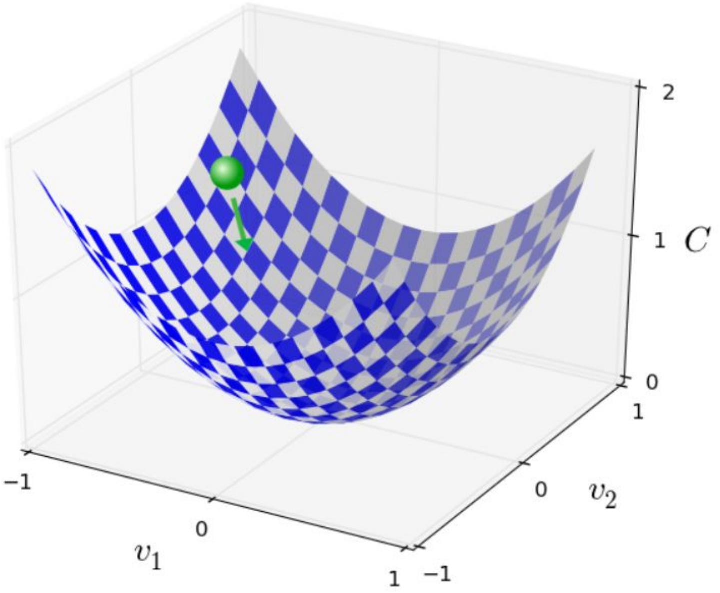
Gradient Descent

- Just move in the opposite direction of the gradient, duh

$$\Delta C \approx \frac{\partial C}{\partial v_1} \Delta v_1 + \frac{\partial C}{\partial v_2} \Delta v_2.$$

$$\Delta C \approx \nabla C \cdot \Delta v.$$

$$\Delta v = -\eta \nabla C,$$



- Stochastic Gradient Descent: update one batch at once
- Other optimisers: Momentum, RMSProp (reading assignment)

$$\boxed{a^l = \sigma(w^l a^{l-1} + b^l).}$$

Backpropagation

1) Calculate the **error** (new term alert) for all layers

$$\boxed{\delta^l_j \equiv \frac{\partial C}{\partial z^l_j}.}$$

2) Use the error to get gradients of weights and biases

$$\frac{\partial C}{\partial b^l_j} = \delta^l_j. \quad \rightarrow \quad \frac{\partial C}{\partial b} = \delta,$$

$$\frac{\partial C}{\partial w^l_{jk}} = a^{l-1}_k \delta^l_j.$$

3) Apply gradient descent!

Backpropagation - 4 equations

Summary: the equations of backpropagation

$$\delta^L = \nabla_a C \odot \sigma'(z^L) \qquad \text{(BP1)}$$

$$\delta^l = ((w^{l+1})^T \delta^{l+1}) \odot \sigma'(z^l) \qquad \text{(BP2)}$$

$$\frac{\partial C}{\partial b_j^l} = \delta_j^l \qquad \text{(BP3)}$$

$$\frac{\partial C}{\partial w_{jk}^l} = a_k^{l-1} \delta_j^l \qquad \text{(BP4)}$$

[Proof for BP2]

Gradients via recursion

It doesn't need to be so complex

We can break down any expression into simple operations and use chain rule to find the gradient of anything.

Code implementation requires **Object Oriented Programming** → we create a Value class to store gradients

ASSIGNMENT

Implement a class ValueMatrix, that instead of storing numbers stores a matrix (as numpy ndarray)

Additional operation → matrix multiplication (@)

Try to make the class **inherit** the ndarray class

1. Intro

Choose one approach to grab the audience's attention right from the start: unexpected, emotional, or simple.

➔ **Unexpected**
Highlight what's new, unusual, or surprising.

➔ **Emotional**
Give people a reason to care.

➔ **Simple**
Provide a simple unifying message for what is to come

How many languages do you need to know to communicate with the rest of the world?

Tip

In this example, we're leading off with something **unexpected.**

While the audience is trying to come up with a number, we'll surprise them with the next slide.

Just one! Your own.

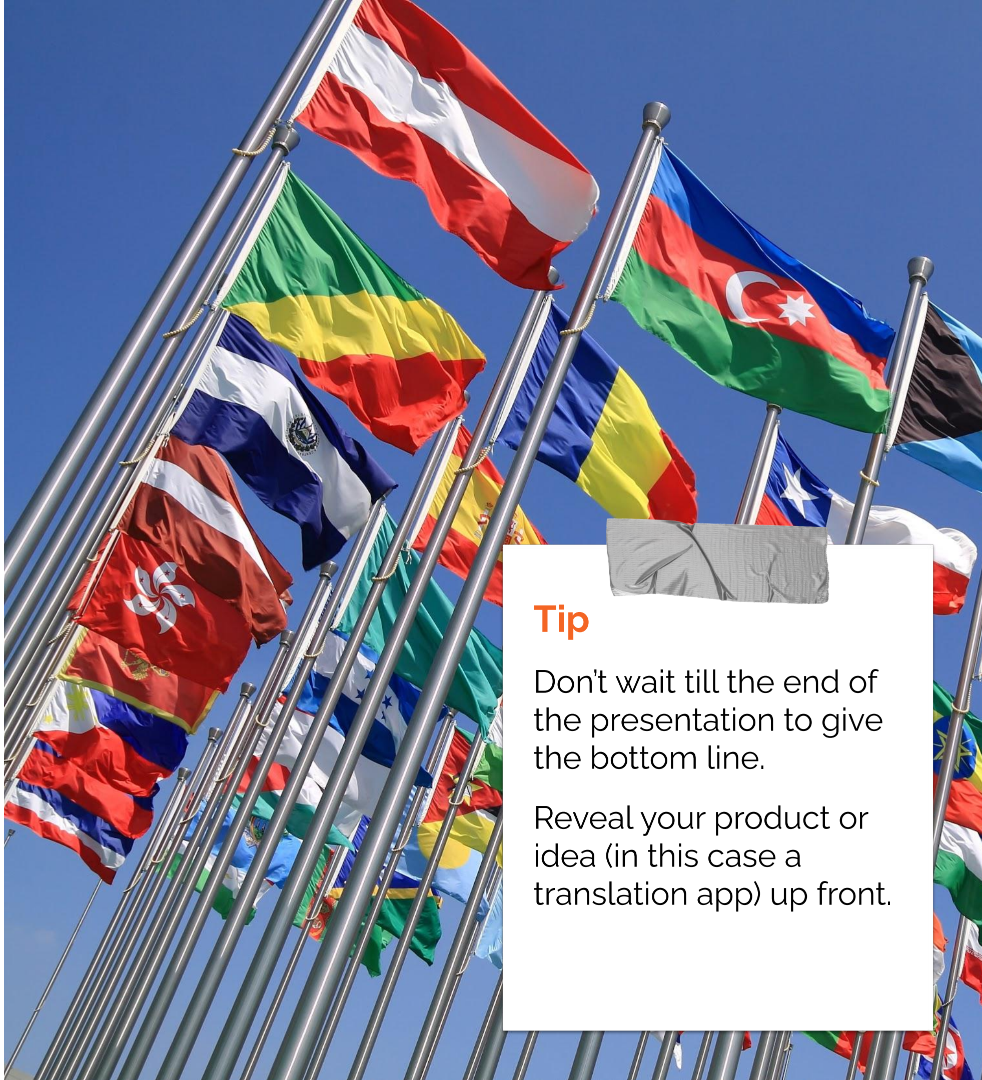
(With a little help from your smart phone)

Tip

Remember. If something sounds like common sense, people will ignore it.

Highlight what is unexpected about your topic.

The Google Translate app can repeat anything you say in up to **NINETY LANGUAGES** from German and Japanese to Czech and Zulu



Tip

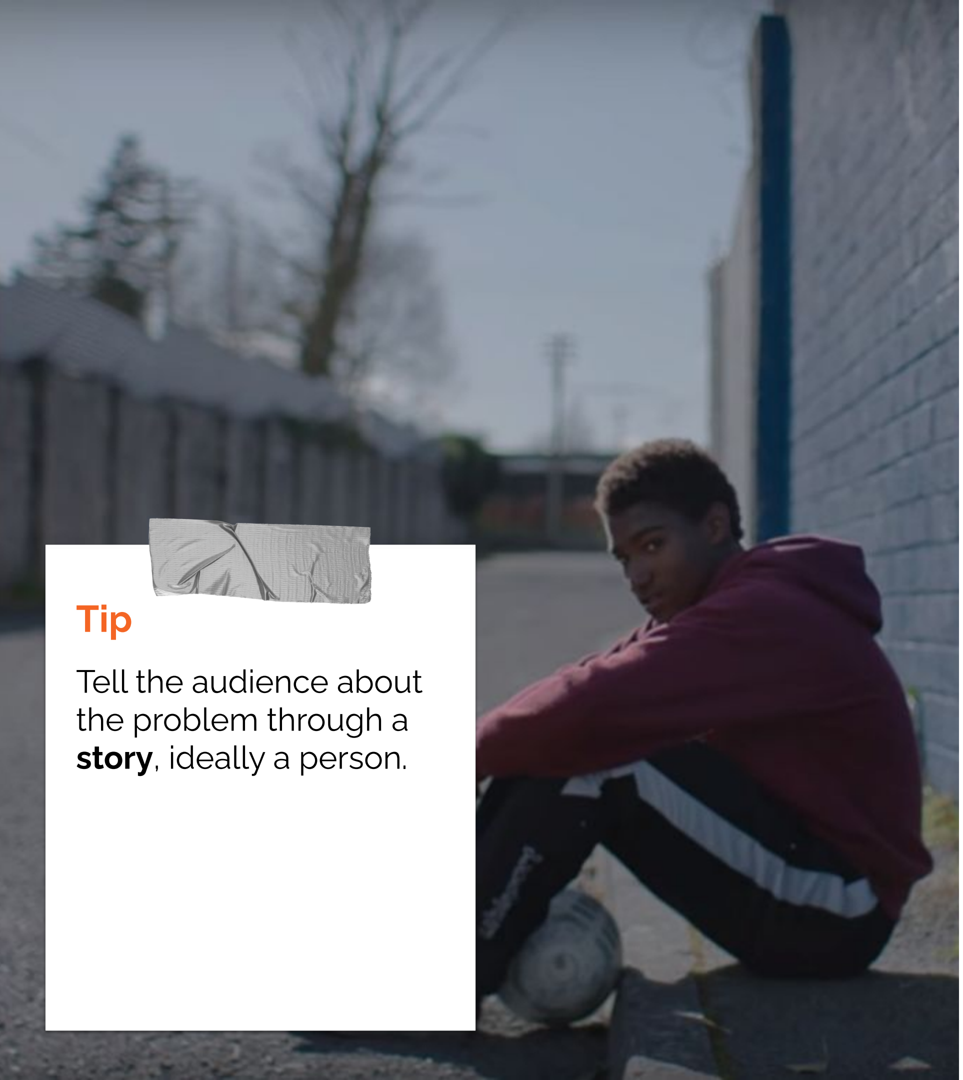
Don't wait till the end of the presentation to give the bottom line.

Reveal your product or idea (in this case a translation app) up front.

2. Examples

By the end of this section, your audience should be able to visualize:

- ➔ **What**
  What is the pain you cure with your solution?
- ➔ **Who**
  Show them a specific person who would benefit from your solution.



Tip

Tell the audience about the problem through a **story**, ideally a person.

Meet Alberto.

He recently moved from Spain to a small town in Northern Ireland.

He loved soccer, but feared he had no way to talk to a coach or teammates.

Meet Marcos.

He recently opened a camera shop near the Louvre in Paris.

Visitors to his store, mostly tourists, speak many different languages making anything beyond a simple transaction a challenge.

Story for illustration purposes only

Tip

If one example isn't sufficient to help people understand the breadth of your idea, pick a couple of examples.

A translation barrier left Alberto feeling lonely and hurt Marco’s business.

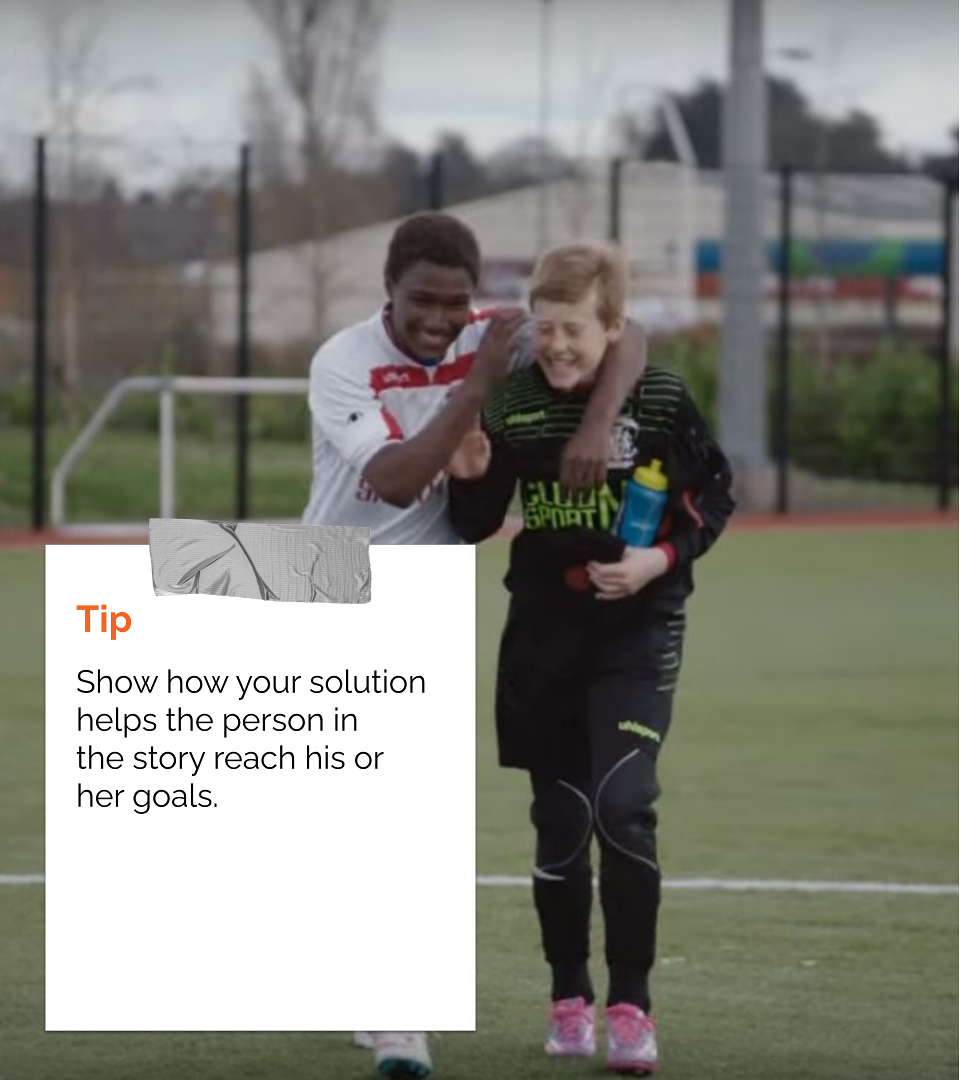
Tip

Ideally, speak of people in very different situations, but where each could benefit from your solution.

Then, Marcos discovered Google Translate

He has his visiting customers speak their camera issues into the app.

He's able to give them a friendly, personalized experience by understanding exactly what they need.



Tip

Show how your solution helps the person in the story reach his or her goals.

A simple gesture

Coaches Gary and Glen knew no Spanish.

They used Google Translate to invite Alberto to join in... “Do you want to play?”... “Can you defend the left side?”

From outsider to star

Alberto scored 30 goals in 21 games. He is now being scouted by several professional clubs in the Premier League. And he's a favorite of the other boys on the team.

[See a short video on Alberto's story]

Tip

Stories become more credible when they use concrete details such as the specific complex moves Alberto learned through Translate and his 30 goals in 21 games performance stats.

3. Examples

People need to understand how rare or frequent your examples are.

Pick 1 or 2 statistics and make them as concrete as possible. Stats are generally not sticky, but here are a few tactics:

- ➔ **Relate**
  Deliver data within the context of a story you've already told
- ➔ **Compare**
  Make big numbers digestible by putting them in the context of something familiar

It's no surprise Marcos uses Google Translate in his shop regularly.

There are 23 officially recognized languages in the EU.

Source: theguardian.com

Tip

Don't let data stand alone. Always relate it back to a story you've already told, in this case, Marco's shop.

More than 50 million Americans travelled abroad in 2015

THAT'S MORE THAN THE POPULATION OF CALIFORNIA AND TEXAS COMBINED

Tip

When a number is too large or too small to easily comprehend, clarify it with a comparison to something familiar.

Source: travel.trade.gov

4. Closing

Build confidence around your product or idea by including at least one of the these slides:

- ➔ **Milestones**
  What has been accomplished and what might be left to tackle?
- ➔ **Testimonials**
  Who supports your idea (or doesn't)?
- ➔ **What's next?**
  How can the audience get involved or find out more?

Milestones

2014

2015

October 2014

Translate web pages with Chrome extension

August 2015

Translate conversations through your Android watch

October 2015

Translate text within an app

November 2015

Translate written text from English or German to Arabic with the click of a camera

What people are saying

With this app, I'm confident to plan a trip to rural Vietnam

Wendy Writer, CA

Visual translation feels like magic

Ronny Reader, NYC

Translate has officially inspired me to learn French

Abby Author, NYC

Quotes for illustration purposes only

Know a 2nd language? Make Google Translate even better by joining the community.

Tip

Inspire your audience to act on the information they just learned.

Depending on your idea, this can be anything from downloading an app to joining an organization.

Good luck!

We hope you'll use these tips to go out and deliver a memorable pitch for your product or service!

For more (free) presentation tips relevant to other types of messages, go to heathbrothers.com/presentations

For more about making your ideas stick with others, check out our book!